Saketh Korada

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EDUCATION

University of California, San Diego (UCSD) **Sep 2024 - June 2027**
B.S. in Computer Engineering *GPA: 4.0/4.0*

Awards/Honors: Provost Honors, Deans List

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Computer Architecture & Organization, Linear Algebra, Discrete Math, Probability & Statistics, Graph Theory

EXPERIENCE

Salesforce | *Software Engineer Intern* Jan 2026 – Present

- Incoming SWE intern on the Hyperforce Platform Services (HPS) Cloud Platform.

Existential Robotics Laboratory (ERL) @ UCSD | *Undergraduate Robotics Researcher* Dec 2025 – Present

- Developing LiDAR-based mapping and autonomous motion-planning algorithms in Python using ROS, Gazebo, and PyBullet.

SEDS @ UCSD | *Software Engineer* Sept 2024 – Present

- Architected a real-time C++ control system with 100 Hz deterministic execution; optimized PID and Kalman filter algorithms to process asynchronous sensor streams with sub-1-degree precision.
- Designed a distributed communication framework across dual microcontrollers and an onboard PC, implementing high-throughput data pipelines via CAN (1 Mbps) and SPI (4 MHz) for telemetry logging.
- Streamlined collaborative development using Git/GitHub and modular firmware design, improving reliability through rigorous code reviews and hardware-in-the-loop testing.

Forsys Inc | *Software Engineer Intern* May 2023 – Sept 2023

- Developed a Python LLM assistant that converted natural-language questions into Salesforce API queries, reducing manual CRM lookups by 90% and increasing team productivity by 30%.
- Built a Salesforce-to-PostgreSQL ETL pipeline with batched loads, indexing, and automated health checks, reducing query latency from minutes to seconds.

PROJECTS

CompressBench | *C++, Go, AWS, Compression Algorithms, Binary I/O, REST APIs* April 2026

- Developed a modular C++ compression benchmarking platform using bit-level streams, binary serialization, custom file headers, and extensible codec interfaces for lossless compression across text, binary, and raw image formats.
- Implemented benchmark tooling to measure compression ratio, space savings, throughput, compression/decompression latency, header and payload size across 100+ test files, reaching up to 65% compression savings and 50+ MB/s throughput on structured data.
- Designed a lightweight Go REST/cloud backend for upload, compress, decompress, verify, benchmark, and download workflows, enabling asynchronous compression jobs, JSON performance reports, and AWS EC2/S3 deployment for large-file processing

Traffic Analytics Platform | *AWS, Docker, Python, YOLOv8* Sept 2025

- Built an AWS-based real-time computer vision pipeline using containerized YOLOv8 with FastAPI and Docker; used multithreading to raise throughput to 120+ FPS with 105 ms P90 latency.
- Designed a serverless analytics layer using AWS Glue, Athena, and Streamlit to query GB-scale data and visualize detection results interactively.

Project Jarvis | *Python, Gemini CLI, REST APIs, SQLite* July 2025

- Extended Gemini CLI via MCP to orchestrate Gmail, Calendar, and Spotify APIs, automating 5+ multi-step workflows including meeting scheduling and playlist generation.
- Integrated OAuth 2.0 authentication and designed a SQLite-based long-term memory layer to maintain 10K+ conversation contexts while achieving sub-second response times in 100+ daily interactions.

Val Gazer | *Python, OpenCV, MediaPipe, Scikit-learn, JavaScript, Overwolf API* May 2025

- Developed a Python-based gaze tracker using OpenCV and MediaPipe (468 facial landmarks at 30 FPS); trained a Random Forest regressor on 5K samples and achieved 97% gaze accuracy.
- Integrated a Python gaze-tracking service with Overwolf's API to process 200+ Valorant events per match and overlay real-time gaze heatmaps for gameplay analysis.

SKILLS

Languages: Python, Java, C/C++, GO (Golang), JavaScript, SQL, MATLAB
Tools: Git/GitHub, Linux, Docker, AWS, VS Code, Codex, PostgreSQL, SQLite
Frameworks/Libraries: FastAPI, OpenCV, MediaPipe, TensorFlow, YOLOv8, ROS, PyBullet
Concepts: Distributed Systems, REST APIs, ETL Pipelines, Binary I/O, Compression Algorithms, Data Structures, Control Systems, Embedded Systems, Robotics, Computer Vision, Agentic Coding, AI-Assisted Development